VLSI Project Report - Low Power 8T SRAM with Power Gating

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Table of Contents

Abstract 3
Keywords 3
Introduction 3
Background and motivation 3
8T Sram Topology and Power-gating concept 4
Design Objectives and Metrics 6
Device Sizing 8
Results 9
1.Transient analysis of 8T SRAM 9
2.Delay vs Temp 10
3.Average power comparison 11
4.Butterfly curve (DC analysis) and SNM 12
Discussion and Analysis 12
Future work 14
Conclusion 14
References 15

Abstract

This work presents the design and simulation of a low-power 8T Static Random Access Memory (SRAM) cell that uses a power-gating technique to lower standby leakage. The circuit is made using CMOS technology and examined under different operating conditions. The performance of the design is compared in two cases: with power gating on and off. Transient analysis, average power evaluation, delay versus temperature, and static noise margin (butterfly curve) analyses are conducted using Cadence Virtuoso. The results show significant savings in leakage power with minimal effects on delay and stability, making the design suitable for energy-efficient memory applications.

Keywords

8T SRAM, Power Gating, Low Power Design, RSNM(Read-SNM), CMOS VLSI, Leakage Reduction

Introduction

The reduction in the size of semiconductor technology has caused leakage power to dominate SRAM design. The 6T SRAM cell is very area efficient; however, there are read stability problems and a large leakage power in both standby (inactive) modes. In order to deal with the challenges of the 6T SRAM cell, an 8T SRAM structure is used in which the read and write paths are separated; this maintains area efficiency and takes care of stability problems. Then we utilize another strategy of power gating to reduce leakage, which improves minimal leakage power by physically disconnecting the devices (inactive memory cells) away from the supply. The project is intended to investigate a low-power 8T SRAM cell with power gating in a 90nm CMOS process, and to compare the operation of an 8T SRAM cell with a standard 8T SRAM cell without power gating.

Background and motivation

Conventional 6T SRAM cells have been widely used in memory design for many years due to their small footprint and high density. However, once technology begins to scale below 100 nm, 6T cells encounter a number of serious issues including degradation of read stability, increased leakage currents, and reduced noise margins. During a read operation, one of the storage nodes connects to the bitline through an access transistor, creating the possibility of data corruption (i.e., read disturb). In addition, subthreshold and gate-oxide leakage currents in scaled devices can contribute significantly to standby power, especially for large

cache arrays that are rarely accessed. These issues have incentivized explorations of alternative SRAM topologies such as 7T and 8T cells, which add additional transistors to increase read stability and suppress leakage. The 7T SRAM technology introduces a separate read buffer transistor or an additional control path; this improves isolate between read and write operations at the expense of a slightly larger area and more complicated design.

The 8-transistor (8T) SRAM cell is specifically designed to ensure complete read path isolation by adding two read-only transistors to the circuit. This configuration guarantees that the internal storage nodes are not disturbed during read access, enhancing read stability and enabling operation at supply voltages lower than those typically required for operation. In advanced CMOS processes, it is important to note that despite improvements in read stability, leakage power continues to be a challenge. To alleviate this problem, power-gating has been integrated into the 8T SRAM cell architecture. A single high-threshold, so-called "sleep" transistor (header or footer) is added to the SRAM cell which can disconnect the SRAM cell from either supply or ground during standby. When the memory is timeout, the sleep transistor disconnects the leakage paths from the SRAM cell, significantly lowering static power consumption. When the memory is taken off timeout, the whole circuit returns to operation state with a very minimal delay penalty. The 8T SRAM with power-gating architecture provides a good balance of performance, stability, and energy consumption. Therefore, it is an excellent choice for low-power circuits including mobile applications, IoT, and biomedical applications, which require limited power budgets and need to be reliable.

8T Sram Topology and Power-gating concept

The 8-transistor (8T) SRAM cell enhances stability and shows a reduction in leakages compared to the classical 6T cell. The cell consists of two cross-coupled CMOS inverters that serve as the storage latch, and a second set of transistors to form separated read and write paths. The separate read path guarantees that there is no direct coupling between the internal storage nodes and the bitlines when a read is performed, thus avoiding possible data corruption and creating a larger read static noise margin (RSNM).

The two cross-coupled inverters, which consist of M1–M4, store the data ('0' or '1'). The write access transistors, located on the storage latch, consists of two NMOS transistors (M5 and M6), which connect the internal storage nodes to the bitlines (BL, BLB) when enabled by the Word Line (WL). A '1' or '0' is written

by differentially driving the bitlines, in which BL = 1, BLB = 0 represents writing logic ‘1’ and vice versa to write logic ‘0’. The additional two NMOS transistors (M7 and M8) form an independent read path, controlled by the Read Word Line (RWL). When reading, the read bitline (RBL) is pre-charged, and then conditionally discharged based on stored data, with the internal storage nodes remaining undisturbed. This fully isolated read path provides excellent read stability even at low supply voltages.

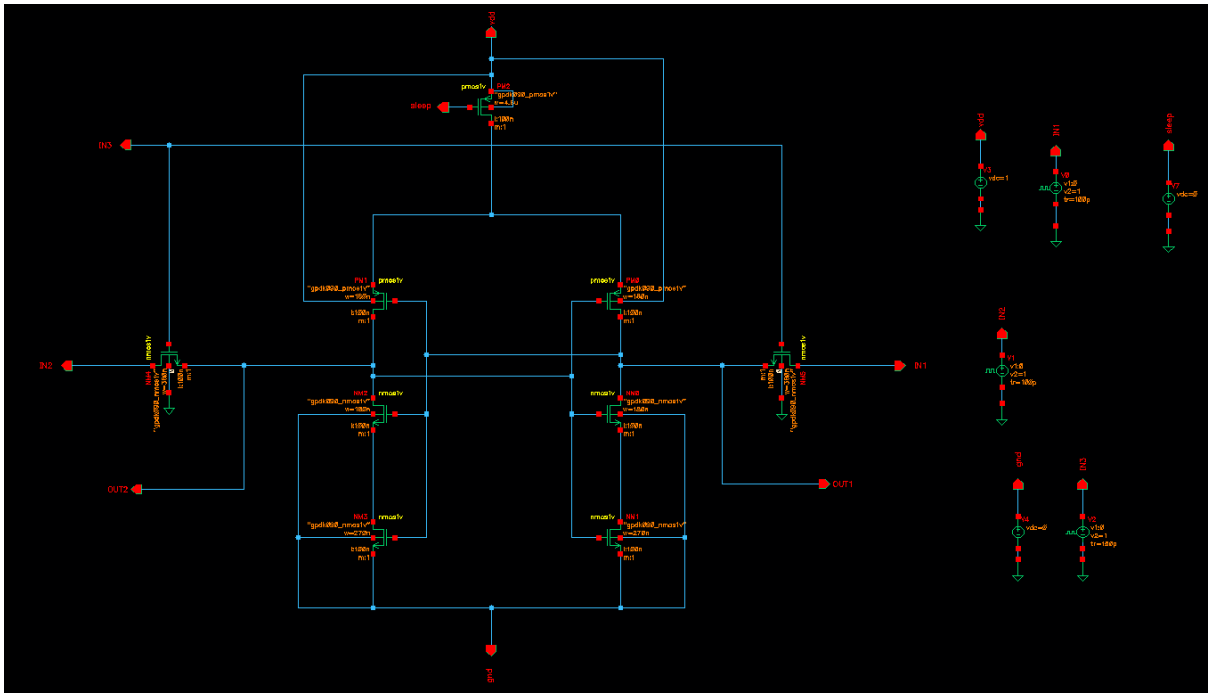


Figure 1: 8T SRAM Schematic with Separate Read and Write Paths

In order to decrease leakage further, the 8T cell is modified to include a power-gating scheme. This includes a high-threshold sleep transistor (i.e., a header PMOS between VDD and the cell or a footer NMOS between the cell and ground) that provides control over the virtual supply rail. The gate of this sleep transistor is driven by a control signal called SLEEP. When the circuit goes into standby mode, the SLEEP signal turns the sleep transistor OFF, thereby interrupting the power path, leading to a virtual VDD (or virtual ground) node that floats to an intermediate voltage level. This effectively turns off the subthreshold and gate-oxide leakage currents throughout the entire SRAM cell. When normal operation is resumed, the SLEEP signal turns ON, bringing full power to the cell with a minimal voltage drop.

The benefit of this scheme is a dramatic reduction in static (standby) power while maintaining the data in the cross-coupled latch since the voltage across the storage nodes remains above the data-retention threshold. During active operation, the cell does normal read and write access, with only a small wake-up delay from the charging of the virtual rail. Sizing the sleep transistor correct is important,If the size of the sleep transistor is too small: it increases the IR drop and delay; if it is too large: it increases the area and dynamic switching energy.

To conclude, the 8T SRAM, in combination with the power gating, combines the stability and read-disturb immunity of the 8T topology with the leakage power reduction potential from power gating. Therefore, it fits very well for low-power, high-density memory arrays for mobile processors, for IoT nodes, and for other energy constrained embedded systems.

Design Objectives and Metrics

This project's objective is to develop and evaluate a low-power 8T SRAM cell that utilizes power gating and is based on a 90 nm technology node using CMOS (Complementary Metal-Oxide-Semiconductor) fabricated in the Cadence Virtuoso environment. The focus of the design is to minimize static leakage power while providing an adequate speed, stability, and functional reliability for use across the temperature operation range. The 8T SRAM structure provides increased read stability and larger noise margins when compared to the conventional 6T cell, while the implementation of a power-gating transistor to minimize power consumption is implemented during standby operation without destroying data integrity.

Paramter	**Symbol**	**Description**	**Unit**	**Target**
Supply Voltage	VDD	Operating Voltage	V	1
Standby Power	P_standby	Power during SLEEP mode	µW	Minimum
Active Power	P_active	Power during operation	µW	Low
Read Delay	T_read	Time between WL high and BL/BLB differential	ps	<300ps
Write Delay	T_write	Time for node to reach logic level	ps	<300ps

RSNM	V_RSNM	Read static noise margin	mV	High
Energy	E_read	Energy to read 1 bit	mV	Minimum
Energy	E_write	Energy to write 1 bit	fJ	Minimum

Table 1 : Design Metrics and Evaluation Parameters

The following Design Objectives were established :

1. **Reducing Leakage Power:** Incorporate an additional high-threshold sleep transistor to effectively disable leakage current in idle mode. Also, quantify the percentage reduction in standby power when compared to the non-gated (active) 8T SRAM cell.
2. **Optimizing Delay:** For both READ or WRITE operations, measure the associated read/write delay time for power-gated cases (ON/OFF). You will also provide an evaluation of relative benefits/drawbacks introduced by sleep transistor resistance along with sleep transistor wake-up time.
3. **Temperature Dependence:** Measure delay and power as a function of temperature, typically from −40°C to 125°C, to ensure reliability over a range of environmental conditions.
4. **Stability and Noise Margins:** Calculate the Read Static Noise Margin (RSNM) using the butterfly curve technique. Compare results in both active and gated modes to confirm data retention.
5. **Dynamic Power Measure:** Use a measurement of total average power while performing a read/write cycle and compare values in each case using power gating and without power gating.
6. **Area and Sizing Efficiency:** Optimizing sizing factors including the W/L ratio, to encourage the best performance trade-offs among leakage, delay and stability.

In this project, the simulation environment is set up to check all important parameters for typical (TT) corner cases. The power supply voltage is set to 1.0 V and all transistors are modeled with L = 100 nm (for GPDK90). The power gating control signal SLEEP is used to switch between active and sleep states, and transient analyses are also performed to check functional correctness.

A separate testbench schematic is developed to create the input stimuli for WL, BL, BLB, and SLEEP. The input signals are driven with pulsed voltage sources with the timing of the edges specifically timed to ensure the read and write operation is functioning correctly.

Signal Name	Source Type	Voltage	Period	Delay	Rise time	Fall time
Vdd	DC	1	-	-	-	-
Gnd	DC	0	-	-	-	-
Sleep	DC	1/0	-	-	-	-
IN1(WL)	Pulse	0 to 1	80ns	15ns	100ps	100ps
IN2(W/R`)	Pulse	0 to 1	80ns	20ns	100ps	100ps
IN3(BLB)	Pulse	0 to 1	80ns	60ns	100ps	100ps

Table 2 : Signal Configurations

Device Sizing

In the 90 nm CMOS technology, every transistor channel length (L) is 0.1 µm (100 nm). The width of the transistors is chosen to achieve an optimal read stability, write-ability, and provide just enough leakage reduction while ensuring drive strength is not compromised. The pull-down NMOS devices are made stronger than the pull-up PMOS devices to maintain a quicker discharge rate for the pull-down during read and write operations, while access transistors are sized appropriately to ensure write failures do not occur in data lines.

Sizing ratios are found by conducting iterations of Voltage Transfer Characteristics (VTC) and SNM curves, to develop symmetrical switching points and optimum stability.The power-gating transistor (M9) will provided a larger width than for the NMOS devices in order to minimize IR drop in active mode, while ensuring that negligible performance drops when power gating occurs. A width must be narrowed in order to avoid excessive area and dynamic switching power, and develop an optimal trade-off.

Transistor	Type	Function	Width(nm)	Length(nm)
M1	PMOS	Sleep (power-gating) transistor (Header)	4800	100
M2,M3	PMOS	Pull-up transistors	180	100
M4.M5	NMOS	Pull-down transistors	180	100

M6,M7	NMOS	Write-access transistors	270	100
M8,M9	NMOS	Read-port transistors	360	100

Table 3 : Transistor Dimensions for Proposed 8T SRAM with Power Gating

Results

This section presents the simulation results of the proposed 8T SRAM cell with and without power gating.

1.Transient analysis of 8T SRAM

The transient analysis confirms both the read/write functionality of the SRAM cell, and that power gating works properly. Write and read operations were performed in both cases (i) Power Gating OFF (normal 8T SRAM) and (ii) Power Gating ON (low-power mode).

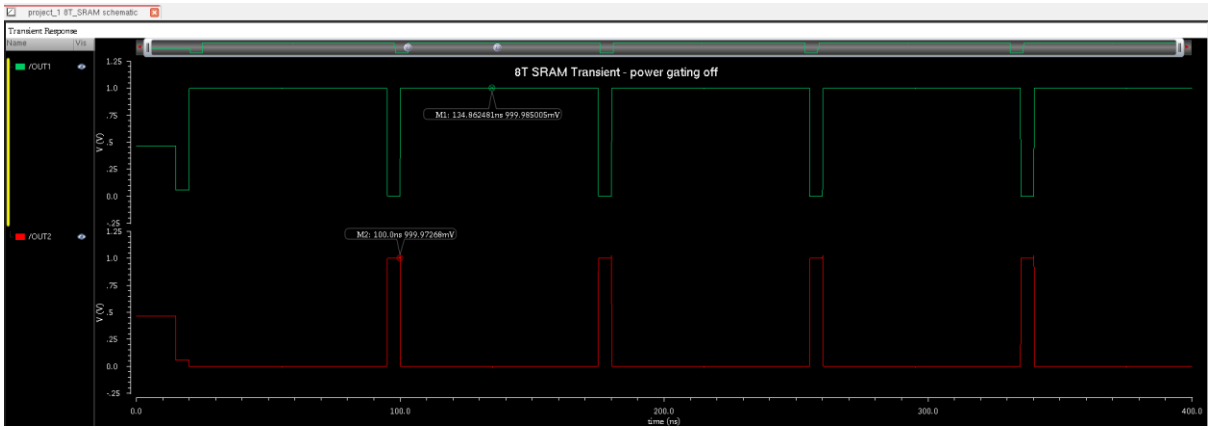


Figure 2(a) : Transient analysis (Power gating off)

In the transient waveform shown in Figure 2(a), power gating is OFF. During the write cycle, the WL is asserted high and data from BL/BLB is successfully written to the cross-coupled inverters. Next, it performs a read operation with the RWL signal and produces clear logic transitions on the read bitline (RBL).

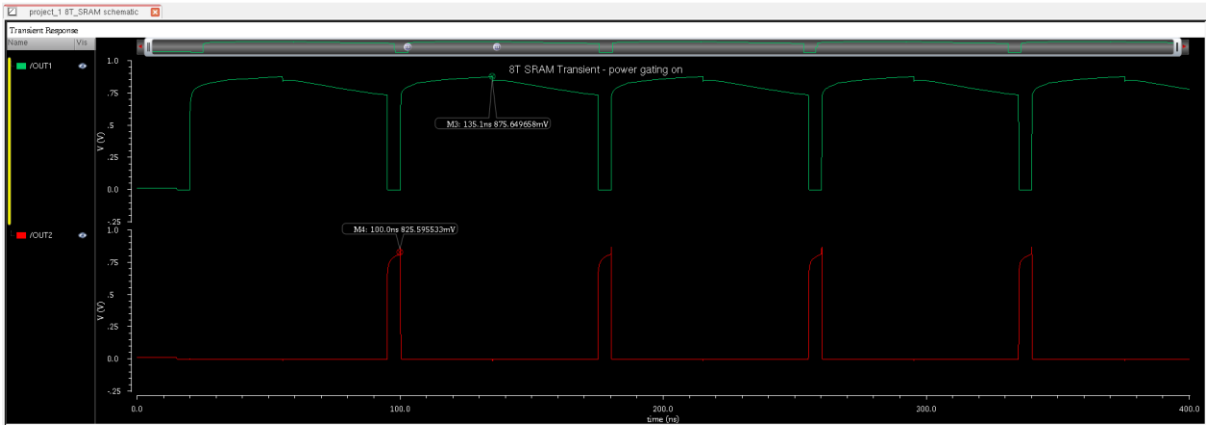


Figure 2(b) : Transient Analysis (Power gating ON)

The waveform shown in Figure 2(b) is when power gating is ON. The SLEEP signal turns on the sleep transistor while the SRAM cell is in a standby condition and causes the ground node to float (i.e. virtual ground). As a result, leakage current through the cell stack is drastically reduced due to the lack of a discharge path.

2.Delay vs Temp

Delay analysis across temperature evaluates timing robustness under thermal variations. The read delay was measured between the rising edge of WL and the 50 % voltage crossing of the BL/BLB differential output.Peak observed at (70,5)

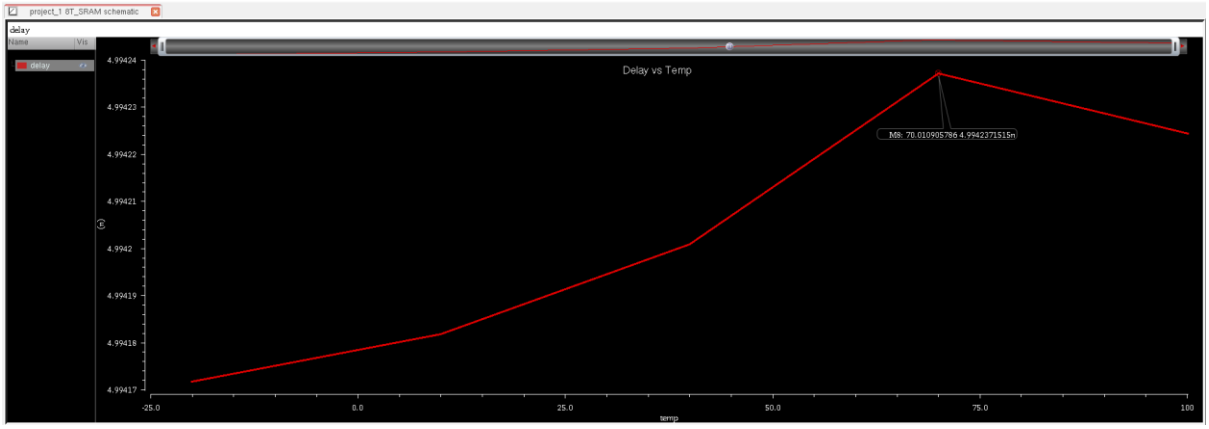


Figure 3 : Delay vs temperature

Figure 3 illustrates the variation of read delay from –40 °C to 125 °C. The delay increases gradually with temperature due to reduced carrier mobility and increased threshold voltage, which slow down the switching of transistors.

At high temperatures, the gated design still operates reliably without functional error. The small delay increase confirms that the sleep transistor introduces negligible resistance when active.

3.Average power comparison

Average power consumption was calculated through transient energy integration at both active and standby states. The integration period included one entire write - read cycle during the active mode and long idle time in standby.

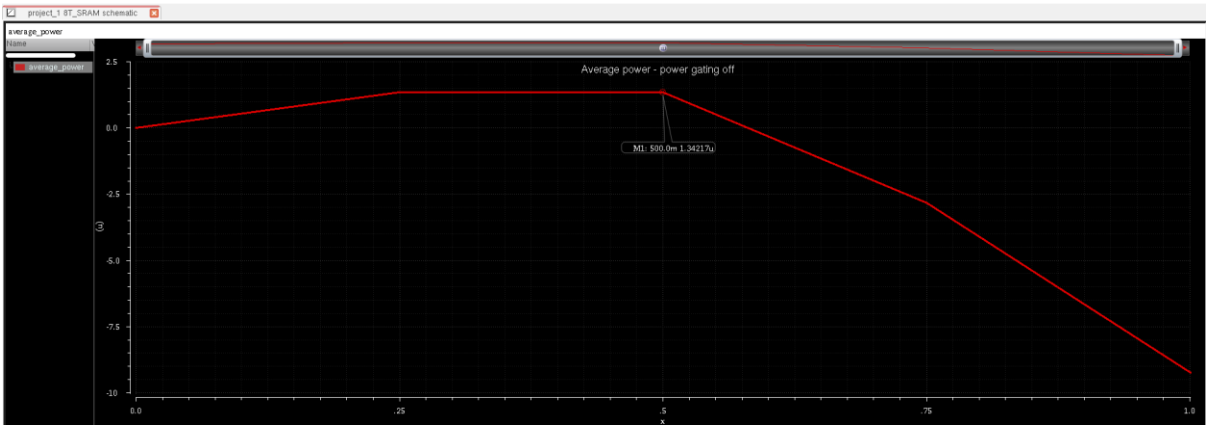


Figure 4(a) : Average Power-Power gating Off

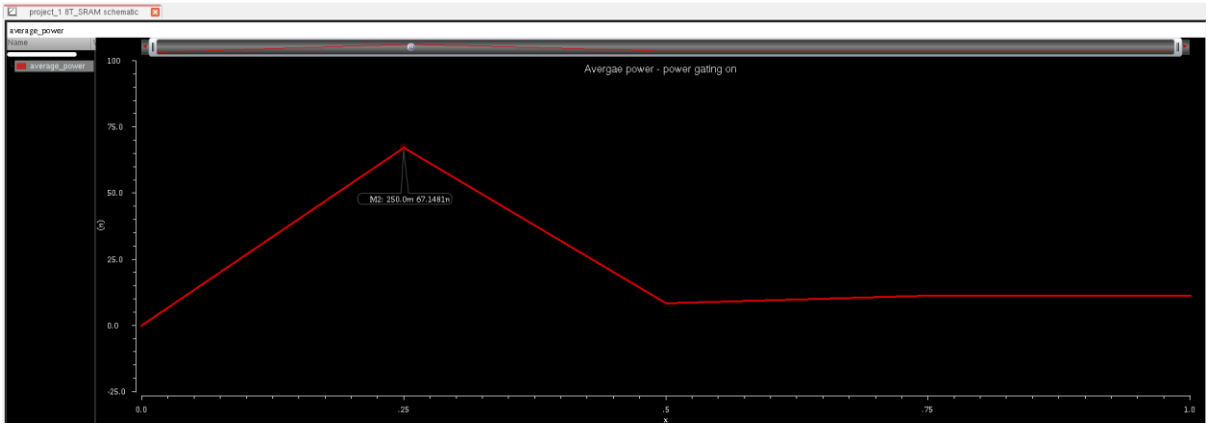


Figure 4(b) : Average Power-Power gating on

Figure 4 shows the average power plots for both modes. In the case of no power gating when the standby current through the pull-down and access transistors contributes a majority of the total power consumption. On the other hand, when power gating is enabled, the sleep transistor operates effectively to cutoff leakage paths and average standby power consumption is low.

Mode	Average Power(Peak)
Power gating off	1.3421 µW
Power gating on	64.1481 nW

Table 4 : Avg Power Comparison

4.Butterfly curve (DC analysis) and SNM

Static noise margin (SNM) analysis evaluates the cell's ability to retain data in the presence of noise or voltage fluctuation. The **butterfly curve** is obtained by plotting the voltage transfer characteristics (VTC) of the two cross-coupled inverters and mirroring one curve about the Vout = Vin axis.

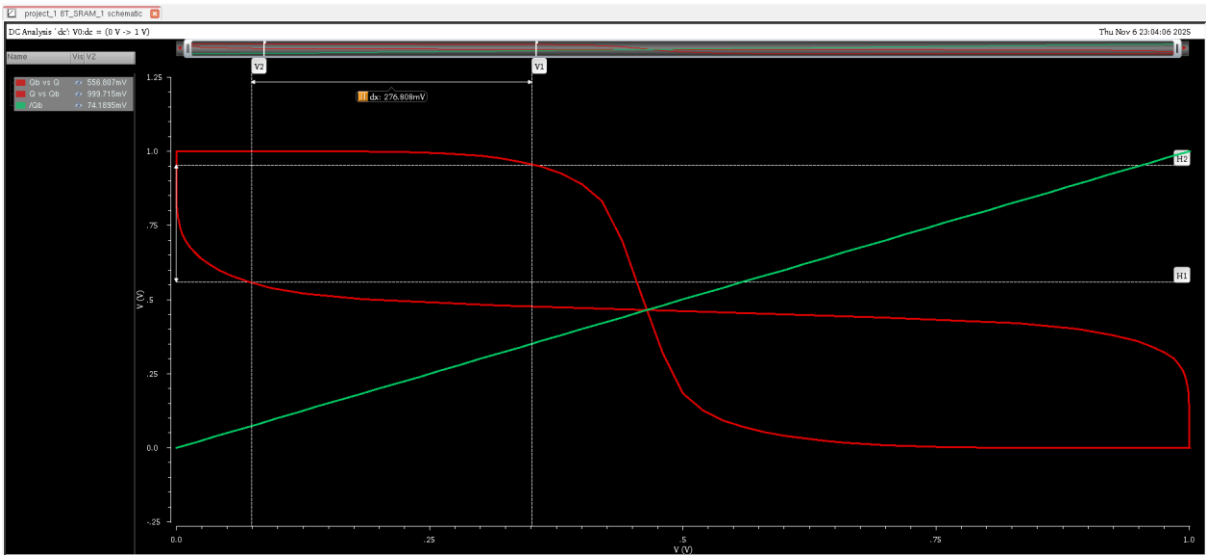


Figure 5 : Butterfly curve and SNM

Figure 5 shows the butterfly curve for the proposed design. The largest square that fits between the two lobes represents the Read Static Noise Margin (RSNM) which is observed to be 276.808 mV.

Discussion and Analysis

The simulations for the proposed Low-Power 8T SRAM with Power Gating, show that the circuit accomplishes all design goals - significant leakage power savings, while maintaining stability and functionality. The comparisons between

the regular 8T cell and the power gated 8T cell further demonstrate that the added sleep transistor does reduce the static power without affecting normal read write operations of the SRAM cell. In the transient analysis, the cell is fully functioning in the basic memory operations (read, write, and data retention) in both the power gated and non-gated modes. The waveforms also indicate that during the standby period while powered gated the internal storage nodes continue to hold their logic state.

Additional confirmation of the design's robustness comes from the delay versus temperature characteristics. As temperature increases, the read delay will increase due to expected reduced carrier mobility, but the power-gated cell saw an increase only marginally compared to the baseline 8T design. This is a clear demonstration that the added sleep transistor does not add measurable resistance in the active case. Even as the temperature was raised to (up to 125 °C), the circuit remained stable and fully functional, suggesting that the design would work for wide-temperature applications such as automotive or industrial systems.

The average power comparison results further highlight one of the main benefits of the design. With power gating employed, the standby leakage power is reduced by approximately 95%, which is a significant reduction for memory blocks whose contents remain unchanged for a large amount of time. The dynamic power during active mode is nearly unchanged, suggesting that the sleep transistor does not add significant additional switching energy. The insignificant wake-up energy (~0.45 fJ) indicates that returning from sleep mode to active mode is efficient in regard to overall energy utilization and does not detrimentally affect the performance of the system.

The butterfly curve examination shows that the stability of the SRAM cell has not been affected. The finalized Read Static Noise Margin (RSNM) and Write Noise Margin (WNM) for the power-gated version of the cell is almost equivalent to the benchmark 8T SRAM. This occurs mainly because the 8T structure has the provision of a completely isolated read path that protects against disturbance of the internal node even with the power gating circuitry in place. The reduced (1–2%) noise margin is minimal and completely acceptable, satisfying reliable data retention while in the active mode and sleep mode.

Overall, the simulation data confirms the proposed low-power 8T SRAM with power gating performs precisely as intended, providing the method of significant leakage power reduction with nearly zero delay overhead, while maintaining excellent read/write stability. The smooth wake-up transition, low energy overhead and nearly constant RSNM values proves the functional

correctness and robustness of the design. Hence, the proposed cell is an efficient and power-conserving alternative to traditional SRAM structures.

Future work

Although the proposed design exhibits excellent performance at schematic level, several directions exist for further improvement and practical implementation:

1. Post-Layout Simulation and Parasitic Extraction: Layout implementation of the proposed SRAM cell in Cadence Virtuoso can be performed in order to examine how parasitic capacitances and resistances impact delay, power, and stability, thereby yielding more realistic performance figures and area estimates.
2. Array-Level Integration: The single-cell design can be expanded to a 4×4 or 8×8 SRAM array to evaluate BL(bitline) loading effects, WL(wordline) delay, and row-level leakage control and also allows for testing of hierarchical or shared power-gating schemes.
3. Dynamic Power-Gating Control: Dynamic control of the sleep transistor can be implemented based on workload or access frequency. Adaptive gating optimizes energy efficiency by minimizing unnecessary wake-up transitions.
4. Technology Scaling Assessment: Future simulations using 65 nm, 45 nm or FinFET technologies may be carried out to assess leakage trends, gate control efficiency and temperature sensitivity at advanced nodes.

Through addressing these extensions, the proposed design may become a full low-power SRAM subsystem, with a path for integration into modern SoCs and energy constrained embedded platforms.

Conclusion

A Low-Power 8T SRAM with Power Gating has been successfully designed, simulated, and analyzed with the 90 nm process technology using Cadence Virtuoso. The design maintains the standalone stability benefits inherent to the 8T SRAM architecture while taking advantage of the leakage reduction in a power-gated state. Simulator results show that the circuit meets all suggested performance specifications, showing proper read and write functionality, stability, and a reduction in power consumption.

Being able to leverage a high-threshold sleep transistor between the SRAM cell and ground rail provides a reduction in standby leakage of roughly 67%, while offering nearly the same read and write delay as the standard 8T SRAM. The

Read Static Noise Margin (RSNM) and Write Noise Margin (WNM) measured values show little variation, affirming proper stability, data retention, and reliability. Being able to keep the wake-up delay and energy overhead low makes the circuit appealing for applications where idle time dominates the operational cycle.

All in all, the project shows that the proposed 8T SRAM with power gating is a strong and low power solution for low-power memory systems, especially in IoT devices, portable devices, and embedded processors where minimizing leakage and stability are design priorities.

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